

STRUCTURA SI ORGANIZAREA CALCULATOARELOR

Calculatoare 3 2025-2026

CURS 4 Sapt 4 21 octombrie 2025 14:00-16:00

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Modul de notare:

Examen:	50%	
Proiect	20%	A3
Laborator:	20%	A2
Test (Midterm):	10%	A1

Obs. Fiecare componenta a notei finale (A1, A2, A3 si Examen) trebuie sa fie promovata cu nota minima 5.

Planificarea activitatilor:

Laborator:

- saptamanal, conform orarului;
- sapt. 13, 14 refaceri laboratoare;

Proiect:

- conform orarului

Curs:

- saptamanal, conform orarului;

Test (Midterm): in saptamana a 11-a (9 decembrie - 13 decembrie)

Continut

Bazele aritmetice ale calculatoarelor electronice

- Sisteme de numeratie

- Reprezentarea numerelor in virgula fixa / mobila

- Operatii aritmetice in virgula fixa / mobila

Proiectarea blocurilor logice pentru operatiile aritmetice in calculatoarele electronice

- Adunarea; scaderea

- Inmultirea

- Impartirea

Proiectarea procesoarelor

- Procesoare RISC

- Setul de instructiuni

- Calea de date (Data Path)

- Calea de control (Control Path)

Bibliografie

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Kaufmann, 2018)

David Harris, Sarah Harris - Digital Design and Computer Architecture
(2007)

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BAZELE ARITMETICE ALE CALCULATOARELOR ELECTRONICE

Adunarea si scaderea in virgula fixa (cod direct)

Operatia: $x \text{ **op** } y = z$

Unde - **op** operatia de realizat, 0 pentru adunare si 1 pentru scadere
 x, y – operanzi, z - rezultat

Operatia finala (pentru semn): $op_{fin} = x_s \oplus y_s \oplus \text{op}$

$$y = a \oplus b$$

a	b	y
0	0	0
0	1	1
1	0	1
1	1	0

Doua cazuri:

1) $opfin = 0 \Rightarrow$ aduna modulele celor doi operanzi, iar semnul rezultatului este dat de semnul primului operand.

Exemplu. $x - y = z$, unde $x = 11$, $y = -14$, pe 6 biti.

$$x + (-y) = z$$

$$[x] = 001011$$

$$[y] = 101110$$

$$opfin = x_s \oplus y_s \oplus op = 0 \oplus 1 \oplus 1 = 0$$

$$\Rightarrow z_s = x_s = 0$$

$$\begin{array}{rcl}
 & & \text{111} \\
 |x| + & & 01011 + \\
 |y| & & 01110 \\
 \hline
 |z| & \Rightarrow & \hline
 & & 11001
 \end{array}$$

CĂZURI:

x_s	y_s	op	$opfin$
0	0	0	0
1	1	0	0
1	0	1	0
0	1	1	0

deci, s-a obtinut $[z] = 011001$ ($z = 25$).

2) $opfin = 1 \Rightarrow$ se scade modulul mai mic din modulul mai mare, iar semnul rezultatului este dat de semnul operandului mai mare in modul.

Exceptie: $op = 1$ (scadere) si $|y| > |x|$ atunci $z_s = \bar{y}_s$.

Exemplu. $x+y=z$, unde $x = -29$, $y = 17$.

$$[x] = 111101$$

$$[y] = 010001$$

$$opfin = x_s \oplus y_s \oplus op = 1 \oplus 0 \oplus 0 = 1$$

x_s	y_s	op	$opfin$
0	0	1	1
0	1	0	1
1	0	0	1
1	1	1	1



$$|x| > |y| \Rightarrow z_s = x_s = 1$$

$$\begin{array}{r} |x| - \\ |y| \\ \hline |z| \end{array} \Rightarrow \begin{array}{r} 11101- \\ 10001 \\ \hline 01100 \end{array}$$

$$\Rightarrow [z] = 101100 \quad (z = -12).$$

$$x = -12 \quad ; \quad y = -17$$

$$x + y = -29 \quad (6 \text{ bits})$$

Code direct
(SM)

$$[x]_d = 101100 \quad ; \quad 12_{(10)} = 1100_{(2)}$$

$$[y]_d = 110001 \quad ; \quad 17_{(10)} = 10001_{(2)}$$

$$\text{op fin} = x_s \oplus y_s \oplus \text{op} = 1 \oplus 1 \oplus 0 = 0 \quad (\text{CA} = 1)$$

$$z_s = x_s$$

$$|x| + |y| = |z|$$

$$\begin{array}{r} |x| \quad 01100 + \\ |y| \quad 10001 \\ \hline 11101 \end{array}$$

$$[z]_d = 111101$$

$$2^4 + 2^3 + 2^2 + 1 = 29$$

Tema Curs SOC Sa se realizeze urmatoarele operatii in cod direct pe **8 biti** folosind ca operanzi numere intregi in baza 10 si urmand regulile prezentate in cursul 2; se vor prezenta detaliat pasii urmati :

1) $-61 + 59$	- + +	$-59 + 61$
2) $-49 + (-34)$	- - +	$-34 + (-49)$
3) $37 - 76$	+ + -	$76 - 37$
4) $-43 - 64$	- + -	$-64 - 43$
5) $-76 - (-87)$	- - -	$-87 - (-76)$
6) $82 + (-73)$	+ - +	$73 + (-82)$
7) $58 - (-49)$	+ - -	$49 - (-58)$

OPF'N
 1
 0
 1
 0
 1
 1
 0

Inmultirea in virgula fixa

(cod direct)

$$x \cdot y = z$$

Etape:

1) *Determinarea semnului rezultatului:*

$$z_s = x_s \oplus y_s$$

2) Calcularea modulului rezultatului:

a) Numere intregi.

$$|x| \cdot |y| = |x| \cdot \sum_{k=0}^{n-2} \underset{\substack{\uparrow \\ \text{poziția} \\ \text{digit}}}{y_k} \cdot 2^k = \sum_{k=0}^{n-2} \underbrace{|x| \cdot y_k \cdot 2^k}_{\text{produs parțial}}$$

$\downarrow$ digit în y $\downarrow$ deplasare

$$|x| \cdot y_k \cdot 2^k = \begin{cases} 0 & \text{dacă } y_k = 0 \\ |x| \cdot 2^k & \text{dacă } y_k = 1 \end{cases}$$

$|x| \cdot 2^k \sim |x|$ deplasat spre stanga k pozitii.

$$\begin{array}{r} 123 \cdot \\ 456 \\ \hline 738 \\ 6150 \\ 49200 \\ \hline 56088 \end{array}$$

$$\begin{array}{r} 101_{(2)} \cdot 5. \\ 110_{(2)} \\ \hline 000 \\ 1010 \\ 10100 \\ \hline 11110_{(2)} \\ 16+8+4+2=30 \end{array}$$

Binar Zecimal

$$11 = 3$$

$$110 = 6$$

$$1100 = 12$$

$$11000 = 24$$

b) Numere subunitare.

0,376589012
 8 ↙ ↘
 9

n-1	n-2	n-3		2	1	0
X _s	X ₋₁	X ₋₂		X _{-m+2}	X _{-m+1}	X _{-m}

↑
 pozitia punctului zecimal
 pentru un numar subunitar

$$|x| \cdot |y| = |x| \cdot \sum_{k=1}^m y_{-k} \cdot 2^{-k} = \sum_{k=1}^m |x| \cdot y_{-k} \cdot 2^{-k}$$

$$|x| \cdot y_{-k} \cdot 2^{-k} = \begin{cases} 0 & \text{daca } y_{-k} = 0 \\ |x| \cdot 2^{-k} & \text{daca } y_{-k} = 1 \end{cases}$$

$|x| \cdot 2^{-k} \sim |x|$ deplasat spre dreapta k pozitii.

0,10110101
 —————→
 0 1

3) Trunchierea si rotunjirea rezultatului (numai pentru numere subunitare).

Exemplu: $x \cdot y = z$ in virgula fixa, $x = \underline{20/32}$ ($=0.625$) si $y = -19/32$ ($=-0.59375$).

$[x] = 0.10100$ ✓
 $[y] = 1.10011$ ✓

$\frac{20}{32} = .10100$

$\frac{19}{32} = .10011$

Etape:

1) Semnul rezultatului: $z_s = x_s \oplus y_s = 0 \oplus 1 = 1$

$\frac{20}{32} = \frac{16 + 4}{32} = \frac{2^4 + 2^2}{2^5} = 2^{-1} + 2^{-3} \rightarrow 0.10100$

$\frac{19}{32} = \frac{16 + 2 + 1}{32} = \frac{2^4 + 2^1 + 2^0}{2^5} = 2^{-1} + 2^{-4} + 2^{-5} \rightarrow 0.10011$

2) Modulul rezultatului:

$$|z| = |x| \cdot |y|$$

$\begin{array}{r} .10100 \bullet \\ .10011 \\ \hline .0000010100+ \\ .0000101000 \\ .0000000000 \\ .0000000000 \\ .0101000000 \\ \hline .0101111100 \end{array}$	$\begin{array}{l} x \cdot y_{-5} \cdot 2^{-5} = x \cdot 2^{-5} \\ x \cdot y_{-4} \cdot 2^{-4} = x \cdot 2^{-4} \\ x \cdot y_{-3} \cdot 2^{-3} = 0 \\ x \cdot y_{-2} \cdot 2^{-2} = 0 \\ x \cdot y_{-1} \cdot 2^{-1} = x \cdot 2^{-1} \end{array}$
--	---

Rezultat exact!

$$\Rightarrow [z] = 1.0101111100$$

$$\begin{aligned} z &= - (0 \cdot 2^{-1} + 1 \cdot 2^{-2} + 0 \cdot 2^{-3} + 1 \cdot 2^{-4} + 1 \cdot 2^{-5} + 1 \cdot 2^{-6} + 1 \cdot 2^{-7} + 1 \cdot 2^{-8} + 0 \cdot 2^{-9} + 0 \cdot 2^{-10}) = \\ &= - (0 \cdot 2^9 + 1 \cdot 2^8 + 0 \cdot 2^7 + 1 \cdot 2^6 + 1 \cdot 2^5 + 1 \cdot 2^4 + 1 \cdot 2^3 + 1 \cdot 2^2 + 0 \cdot 2^1 + 0 \cdot 2^0) / 2^{10} = \\ &= - (256 + 64 + 32 + 16 + 8 + 4) / 1024 = - 380 / 1024 \end{aligned}$$

$= 0,37109375$

(corect !)

-1 -2 -3

$$0, 01100 =$$

$$= 2^{-2} + 2^{-3} =$$

$$= \frac{(2^{-2} + 2^{-3}) 2^5}{2^5} =$$

$$= \frac{2^3 + 2^2}{2^5} = \frac{8 + 4}{32} = \frac{12}{32}$$

3) Trunchiere si rotunjire:

$$\begin{array}{r} .01011 \ 11100 \\ \downarrow \\ .01011 + \\ 1 \\ \hline .01100 \end{array}$$

Rezultatul aproximativ $[z] = 1.01100 \Rightarrow z = \underline{\underline{-12/32}}$ (s-a obtinut rezultatul -0.375, fata de cel exact -0.37109375).

$$6 - 2^6 = 64$$

$$\frac{14}{64} = \frac{2^0 + 2^1 + 2^4}{2^6} =$$

$$= 2^{-6} + 2^{-5} + 2^{-2} \rightarrow$$

$$0,010011\text{ (2)}$$

$$\frac{14}{64} = 0,21875_{(10)} \times 2$$

$$0,53750$$

$$0,53750_{(10)} \times 2 =$$

$$1,07500$$

END 14 oct '25

$$0,582 \quad 0,196500_{(10)} \times 2 =$$

$$0,393000$$

$$0,587 \quad 0,393 \times 2 = 0,786$$

$$0,59 \quad 0,786 \times 2 = 1,572$$

$$0,59 \quad 1,572 \times 2 = 3,144$$

$$0,156 \times 2 = 0,312$$

$$0,312 \times 2 = 0,624$$

$$0,624 \times 2 = 1,248$$

Impartirea in virgula fixa (cod direct)

Metoda comparatiei (numere subunitare)

Se trateaza separat semnele si separat modulele operanzilor.

Conditie: deimpartitul $<$ impartitorul.

$x : y \Rightarrow$ catul q si restul r . (Conditii: $0 < |x|; |y| < 1$, $|x| < |y|$)
(cei $n-1$ biti ai modulului fiecarui operand se noteaza cu indici negativi:

$-1, -2, -3, \dots -m$ ($m = n-1$),

pentru ai pune in corespondenta cu puterile negative ale bazei 2 reprezentand ponderile).

Algoritmul:

```

citeste x,y,m;      m - este precizia dorita (aproximarea)
daca |x| ≥ |y| atunci
|      scrie „Eroare!”
altfel
|       $q_s = x_s \oplus y_s$ 
|       $r_s = x_s$ 
|      |r| = |x|      //initializare rest partial
|      pentru i=1,m executa
|      |      |r| = |r| • 2      //deplaseaza o pozitie stanga
|      |      daca |r| ≥ |y| atunci
|      |      |       $q_{-i} = 1$       //bitul curent al catului
|      |      |      |r| = |r| - |y|
|      |      altfel
|      |      |       $q_{-i} = 0$       //bitul curent al catului
|      |      |      |r| = |r|
|      |      |r| = |r| • 2-m      //rest final
scrie q, r

```

Corectie: restul inmultit cu 2^{-m} . Justificare ($r^{(k)}$ restul partial la pasul k):

$$|r^{(0)}| = |x|$$

$$|r^{(1)}| = 2 \cdot |r^{(0)}| - q_{-1} \cdot |y|; \quad 2 \cdot |r^{(0)}| - |r^{(1)}| = q_{-1} \cdot |y|$$

$$|r^{(2)}| = 2 \cdot |r^{(1)}| - q_{-2} \cdot |y|$$

.....

$$|r^{(m)}| = 2 \cdot |r^{(m-1)}| - q_{-m} \cdot |y|$$

unde

$$q_{-k} = \begin{cases} 1 & \text{daca } 2 \cdot |r^{(k-1)}| \geq |y| \\ 0 & \text{altfel} \end{cases}$$

In ultima relatie de mai sus se inmultesc ambii membri cu 2^{-m} , inlocuindu-se succesiv fiecare $r^{(k)}$ din relatia precedenta in functie de $r^{(k-1)}$.

$$\begin{aligned}
 2^{-m} \cdot |r^{(m)}| &= \\
 &= -2^{-m} \cdot q_{-m} \cdot |y| + 2^{-m} \cdot 2 \cdot (2 \cdot \dots \cdot (2 \cdot |x| - q_{-1} \cdot |y|) \dots - q_{-m+1} \cdot |y|) = \\
 &= 2^{-m} \cdot 2^m \cdot |x| - |y| \cdot (q_{-1} \cdot 2^{-1} + q_{-2} \cdot 2^{-2} + \dots + q_{-m} \cdot 2^{-m}) = \\
 &= |x| - |y| \cdot |q|
 \end{aligned}$$

$$\Rightarrow |x| = |y| \cdot |q| + 2^{-m} \cdot |r^{(m)}|$$

rest corect $2^{-m} \cdot |r^{(m)}|$.

Exemplu. $x : y$, $x = 20/32$ si $y = 25/32$.

$$[x]_d = 0.10100$$

$$[y]_d = 0.11001$$

Impartirea se executa in cinci $(n-1)$ pasi $(1,2,\dots,5)$:

$$\frac{20}{32} = \frac{1 \cdot 2^4 + 1 \cdot 2^2}{32} \rightarrow 0.\underline{1}\underline{0}\underline{1}\underline{0}\underline{0}$$

$\rightarrow 2^5$

$$\frac{25}{32} = \frac{1 \cdot 2^4 + 1 \cdot 2^3 + 1 \cdot 2^0}{32} \rightarrow 0.\underline{1}\underline{1}\underline{0}\underline{0}\underline{1}$$

$$0.10100 : 0.11001 \rightarrow 2, 12$$

$$0 < |x| < 1$$

$$0 < |y| < 1$$

$$|x| < |y|$$

Pas	$ y $	$ r^{(k)} $	q	
0	.11001	.10100	$ r^{(0)} < y $	$\Rightarrow$ OK!
1	.11001	1.01000- .11001 .01111	$ r^{(1)} \geq y $	$\Rightarrow q_{-1}=1$
2	.11001	.11110- .11001 .00101	$ r^{(2)} \geq y $	$\Rightarrow q_{-2}=1$
3	.11001	.01010	$ r^{(3)} < y $	$\Rightarrow q_{-3}=0$
4	.11001	.10100	$ r^{(4)} < y $	$\Rightarrow q_{-4}=0$
5	.11001	1.01000- .11001 .01111	$ r^{(5)} \geq y $	$\Rightarrow q_{-5}=1$
$q = 0.11001$ $r = 0.01111$				
Rest care trebuie corectat prin inmultire cu $2^{-m} = 2^{-5}$				

=>

$[q] = 0.11001$ ($q = 25/32$) si $[r] = 0.0000001111$ ($r = 15/1024$).

Verificare: $x = q \cdot y + r$ ($20/32 = 25/32 \cdot 25/32 + 15/1024$).

$$\begin{aligned} 0.11001_{(2)} &= \left(2^{-1} + 2^{-2} + 2^{-5} \right)_b = \frac{(2^{-1} + 2^{-2} + 2^{-5}) \cdot 2^5}{2^5} = \frac{2^4 + 2^3 + 1}{2^5} = \frac{16 + 8 + 1}{32} = \frac{25}{32} \\ 0.0000001111_{(2)} &= \left(2^{-7} + 2^{-8} + 2^{-9} + 2^{-10} \right)_b = \frac{(2^{-7} + 2^{-8} + 2^{-9} + 2^{-10}) \cdot 2^{10}}{2^{10}} = \frac{2^3 + 2^2 + 2 + 1}{2^{10}} = \frac{8 + 4 + 2 + 1}{1024} = \frac{15}{1024} \end{aligned}$$

$$q \cdot y + r = \frac{25}{32} \cdot \frac{25}{32} + \frac{15}{1024} = \frac{625}{1024} + \frac{15}{1024} = \frac{640}{1024} \stackrel{\div 32}{=} \frac{20}{32}$$

$$\begin{array}{r}
 \leftarrow \\
 \hline
 10101 \quad | \quad 110 \\
 \hline
 10101 \quad | \quad 000011 \\
 \hline
 10101 \quad | \quad \dots \quad \boxed{1} \\
 \hline
 10101 \quad | \quad \boxed{1} \\
 \hline
 10101 \quad | \quad 1 \\
 \hline
 110 \\
 \hline
 -100 \quad | \quad 1 \\
 \hline
 1001 \quad | \quad 1 \\
 \hline
 110 \\
 \hline
 011 \quad | \quad \boxed{1} \\
 \hline
 \boxed{1} \\
 \hline
 \text{next}
 \end{array}$$

$$\begin{array}{r}
 \underline{\underline{6943}} \quad | \quad \underline{\underline{71}} \\
 \hline
 \underline{\underline{639}} \quad | \quad \underline{\underline{0097}} \\
 \hline
 -553 \\
 \hline
 497
 \end{array}$$

$$\begin{array}{r}
 \leftarrow \\
 \underline{\underline{6943}} \quad | \quad \underline{\underline{71}} \\
 \hline
 6543 \quad | \quad 00 \\
 \hline
 6943 \quad | \quad 00 \\
 \hline
 6943 \quad | \quad 00
 \end{array}$$

Exemplu suplimentar (metoda comparatiei modificata):

$$x : y \quad [x] = 1.01000_{(2)} = 5/4_{(10)} \quad [y] = 0.11001_{(2)} = 25/32_{(10)} \quad x : y = 8/5$$

$$\frac{5}{4} : \frac{25}{32} = \frac{5}{4} \cdot \frac{32}{25} = \frac{8}{5}$$

Operatia care trebuie efectuata: $[x] = [y] \cdot c + d$ cu c – catul asteptat si d – restul asteptat.

Echivalam operatia folosind alti operanzi obtinuti prin deplasari la dreapta ale operanzilor initiali:

$$[a] = 2^{-1} \cdot [x] = .10100 \quad [b] = [y] = .11001$$

Observam ca la efectuarea operatiei echivalente obtinem: $[a] = [b] \cdot q + r$ sau $2^{-1} \cdot [x] = [y] \cdot q + r$ sau

$$[x] = [y] \cdot q \cdot 2 + r \cdot 2 \quad \text{sau} \quad [x] = [y] \cdot q \cdot 2 + r \cdot 2$$

De unde catul si restul asteptate la operatia ceruta initial sunt : $c = q \cdot 2$ (catul final deplasat stanga), respectiv $d = r \cdot 2$ (restul final corectat! deplasat stanga)

Pas	$ y $	$ r^{(k)} $	q
0	.11001	.10100	$ x = r^{(0)} < y \Rightarrow \text{OK!}$
1	.11001	1.01000- .11001	$ r^{(1)} \geq y \Rightarrow q_{-1}=1$
		.01111	
2	.11001	.11110- .11001	$ r^{(2)} \geq y \Rightarrow q_{-2}=1$
		.00101	
3	.11001	.01010	$ r^{(3)} < y \Rightarrow q_{-3}=0$
4	.11001	.10100	$ r^{(4)} < y \Rightarrow q_{-4}=0$
5	.11001	1.01000- .11001	$ r^{(5)} \geq y \Rightarrow q_{-5}=1$

$\Rightarrow$.01111 Rest care trebuie corectat prin inmultire cu $2^{-m}=2^{-5}$
 $[q] = 0.11001$ ($q = 25/32$) si $[r] = 0.0000001111$ ($r = 15/1024$).

Verificare: $x = q \cdot y + r$ ($20/32 = 25/32 \cdot 25/32 + 15/1024$).

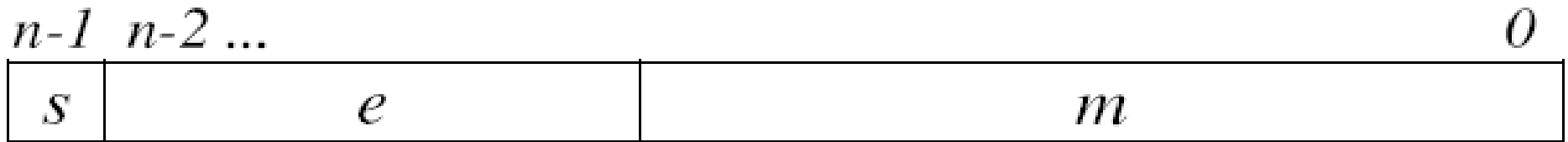
Revenind la catul si restul asteptate initial, obtinem:

$c = 1.1001$, respectiv $d = 0.000001111$ sau $c = \frac{25}{16}$, respectiv $d = \frac{15}{512}$

Verificare: $x = c \cdot y + d$ (~~25/16~~ $\frac{25}{16} \cdot \frac{25}{32} + \frac{15}{512} = \frac{625}{512} + \frac{15}{512} = \frac{640}{512} = \frac{5}{4}$).

$$\frac{25}{16} \cdot \frac{25}{32} + \frac{15}{512} = \frac{625}{512} + \frac{15}{512} = \frac{640}{512} \stackrel{\div 128}{=} \frac{5}{4}$$

Reprezentarea numerelor in virgula mobila



- s bitul de semn al numarului ($s=0$ numar pozitiv si $s=1$ numar negativ);
- m mantisa numarului (cifrele semnificative);
- e exponentul (puterea la care trebuie ridicata o valoare numita baza si care inmulteste cifrele semnificative ale numarului). – se foloseste caracteristica!!!

$$\text{valoare} = (-1)^s \cdot m \cdot \text{baza}^e$$

(baza: 2, 10 sau 16).

Exponentul poate fi negativ, dar in reprezentarea in virgula mobila se foloseste caracteristica – numar pozitiv care nu are nevoie de bit de semn!

$-2.157 \cdot 10^3$
 $\uparrow$
 $0 \div 9$

$20.157 \cdot 10^3$
 $\rightarrow$
 $2.0157 \cdot 10^4$

$0.1010 \cdot 2^{10}$

forma normalizata

parte intregi 0!
 primul digit 1
 (subnormal)

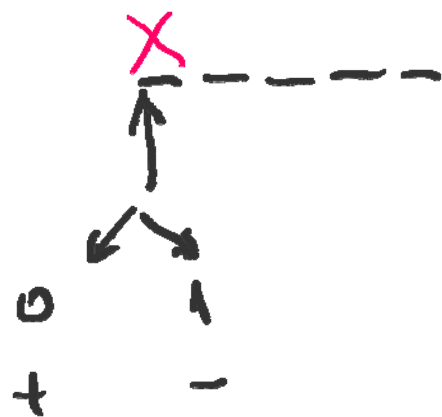
Mantisa:

$1/\text{baza} \leq m < 1$ (baza 2 $\Rightarrow \frac{1}{2} \leq m < 1$ ~ primul bit, c.m.s. este 1).

Exponentul folosit in reprezentare:

caracteristica = exponent + $2^{\text{numar de biti exponent} - 1}$

exponent reprezentat pe 5 biti $2^5 = 32$ $[0; 31]$
 (nr. pozitiv)
 2^{-13} 2^{14} -----> $[-16; +15]$
 (in cu semn)



nr biti exp - 1
 2^{5-1}

$$= 2^{5-1} = 2^4 = 16.$$

$[-16; +15] + 16 \rightarrow [0; 31]$

Caracteristica va
 conduce la un
 exponent non
 totdeauna ≥ 0 !

exponent pe 6 biti $2^6 = 64$
 $[0; 63]$ $[-32; 31]$ $2^{6-1} = 2^5 = 32$
 $\uparrow$ $\underbrace{\hspace{10em}}_{+32}$

Exemplu. 7 biti pentru exponent => 128 valori distincte cu semn: $2^7 = 128$

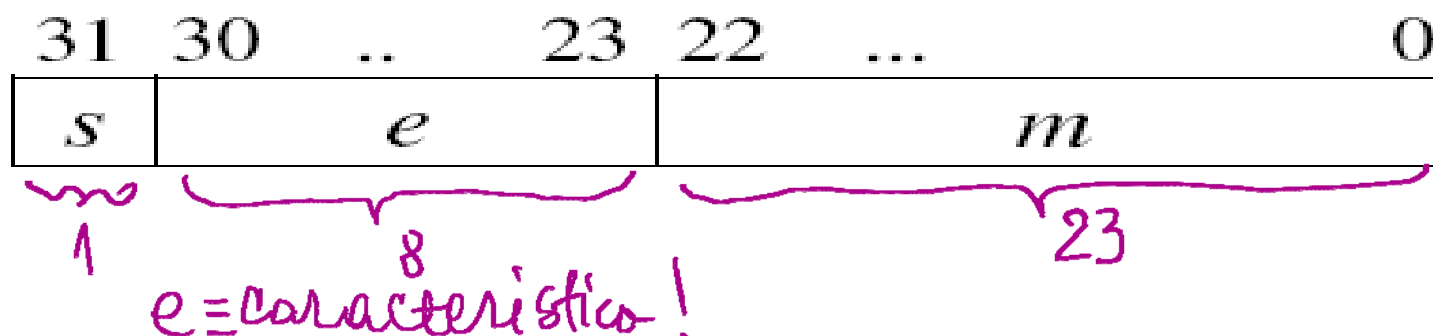
$$-64 \leq \text{exponent} \leq +63$$

$$\text{caracteristica} = \text{exponent} + 2^{7-1}$$

$$\text{caracteristica} = \text{exponent} + 64 \Rightarrow 0 \leq \text{caracteristica} \leq 127$$

EBCDIC IBM
ASCII

Exemplu. Standardul IEEE 754 (IEEE – „Institute of Electrical and Electronics Engineers”) pentru reprezentarea numerelor reale in virgula mobila pe lungime (precizie) simpla (32 de biti):



daca $0 < e < 255$ (caracteristica!) $\Rightarrow$ valoarea = $(-1)^s \cdot 0.m \cdot 2^{e-128}$

daca $e = 0$ si $m = 0 \Rightarrow$ valoarea = 0

daca $e = 0$, $m \neq 0$ sau $e = 255 \Rightarrow$ eroare

caracteristica = exponent + $2^{8-1} = \text{exponent} + 2^7 = \text{exponent}_{\text{real}} + 128$

exponential $\in [-128; 127]$ $\rightarrow$ caracteristica $\in [0; 255]$
 (pe 8 biti) (exponent deplasat)

Exemple de reprezentare

in virgula mobila pe 11 biti (un bit de semn, 5 biti pentru exponent si 5 biti pentru mantisa)

Exponent pe 5 biti => 32 valori distincte cu semn: $-16 \leq \text{exponent} \leq +15$

Caracteristica = exponent + 2^{5-1} = exponent + 2^4 = exponent + 16; $0 \leq \text{caracteristica} \leq 31$

$e_c \text{ max} = 31$; $e_c \text{ min} = 0$;

$$A = (28/32) \cdot 2^8$$

Mantisa $(28/32)_{(10)} = 0.11100_{(2)}$
Semn 0 (+)

Verificare $28/32 = 0.875 = 1/2 + 1/4 + 1/8$

Caracteristica $8 + 2^4 = 8 + 16 = 24_{(10)} = 11000_{(2)}$

Reprezentarea in virgula mobila pentru A:

0 11000 11100 = 31C₍₁₆₎
5 exp depl Mantisa
 (caracteristica)

1C
03

$$B = (30/32) \cdot 2^6$$

Mantisa $(30/32)_{(10)} = 0.11110_{(2)}$
Semn 0 (+)

Verificare $30/32 = 0.9375 = 1/2 + 1/4 + 1/8 + 1/16$

Caracteristica $6 + 2^4 = 6 + 16 = 22_{(10)} = 10110_{(2)}$

Reprezentarea in virgula mobila pentru B:

0 10110 11110 = 2DE₍₁₆₎
5 car. Mantisa

$$C = -\frac{27}{32} \cdot 2^{-10}$$

$$S = 1$$

CHARACTERISTICA
(EXPONENT DEPLASAT)

$$= -10 + 2^4 = -10 + 16 = 6 \rightarrow \underline{06110}_{(2)}$$

$$\text{MANTISA} = \frac{27}{32} = \frac{2^4 + 2^3 + 2^2 + 1}{2^5} = (2^4 + 2^3 + 2^2 + 2^0) \cdot 2^{-5} = (2^{-1} + 2^{-2} + 2^{-4} + 2^{-5})$$

$$= 1 \cdot 2^{-1} + 1 \cdot 2^{-2} + 1 \cdot 2^{-4} + 1 \cdot 2^{-5} \rightarrow 0.11011$$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $2^{-1} \quad 2^{-2} \quad 2^{-4} \quad 2^{-5}$

$$C = \underbrace{100110}_{S \text{ caract.}} \underbrace{11011}_{\text{mantisa}} = 4 \text{ DBh}$$

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$$D = \frac{13}{32} \cdot 2^{-5}$$

$$S = 0 \text{ (nr pozitiv)}$$

$$\text{CHARACTERISTICA} = -5 + 16 = 11 \rightarrow 01011_{(2)}$$

$$\text{MANTISA} = \frac{13}{32} = \frac{2^3 + 2^2 + 2^0}{2^5} = (2^3 + 2^2 + 2^0) \cdot 2^{-5} = 1 \cdot 2^{-2} + 1 \cdot 2^{-3} + 1 \cdot 2^{-5}$$

$$\text{MANTISA} = 0.01101$$

$$\underbrace{0.01101}_{\text{nr este normalizata!}} \cdot 2^{11} \rightarrow 0.11010 \cdot 2^{10}$$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $2^{-1} \quad 2^{-2} \quad 2^{-3} \quad 2^{-4} \quad 2^{-5}$

$$D = \underbrace{001010}_{\text{semn caracteristica}} \underbrace{11010}_{\text{mantisa normalizata}}$$